

Respiratory Acidosis

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 Adapted from: [Respiratory Acidosis](#)

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Learning Objectives (4)

Versions

After completing this brick, you will be able to:

- 1 Differentiate the clinical presentation of respiratory acidosis.
- 2 Explain the pathophysiology of respiratory acidosis.
- 3 Interpret clinical and lab findings to properly diagnose respiratory acidosis.
- 4 Outline the management plan for respiratory acidosis.

CASE CONNECTION

WN is a 22-year-old unresponsive male who is brought into the emergency department by his friend. “I found him on the couch. I couldn’t wake him up and he was breathing slowly! I saw needles on the floor.” On your focused

examination, WN is not responding to his name. He is breathing only 7 times per minute. His pupils are pinpoint, and he has fresh injection sites on his arm.

What is your diagnosis? What will be your immediate first steps? Consider your answers as you read, and we'll revisit WN at the end of the brick.

[GO TO CONCLUSION](#) ↓

What Is Respiratory Acidosis?

The body depends on the pH being within a normal range in order for everything to function appropriately. **The body keeps pH within a narrow range of 7.35 to 7.45.** When the pH falls outside that range, the proteins or machinery of the body change their tertiary structure and function less effectively, leading to a host of problems. The natural tendency of the body is to become acidic. The reason for this is that the Krebs cycle naturally produces carbon dioxide, which gets converted into carbonic acid. The Krebs cycle produces more than 15,000 millimoles of CO₂-derived acid per day. These are called volatile acids. In addition, acids called *non-volatile acids* are unrelated to CO₂ and are excreted by the kidneys. The body maintains acid-base balance by:

- Exhaling CO₂ through the lungs
- Excreting non-volatile acids through the kidneys

Metabolic disorders and respiratory alkalosis are discussed in separate bricks; here, we will focus on respiratory acidosis. **Respiratory acidosis is characterized by hypoventilation or poor gas exchange in the lungs,** leading to:

- Elevated partial pressure of arterial carbon dioxide (PaCO₂) (>40 mm Hg)

- Reduced blood pH (<7.35)

hyperventilation, or due to poor gas exchange in the lung. Respiratory acidosis is an elevation in PaCO_2 caused by

How Do Patients With Respiratory Acidosis Present? (LO1)

Patients with acute respiratory acidosis present with three main categories of symptoms: varying degrees of altered mental status, dyspnea, as well as symptoms from the underlying driver of disease.

Patients with acute respiratory acidosis present with varying degrees of altered mental status, dyspnea, or both. The clinical appearance of the patient depends on a culmination of multiple pathophysiological factors converging into what you see at the bedside, though:

- High levels of CO_2 itself alters normal brain function.

- High CO₂ levels in the brain also drop the pH of the cerebrospinal fluid (CSF), which alters sodium and calcium channels and affects neurotransmitter activity.
- High CO₂ levels in the CSF cause cerebral vasodilation which increased intracranial pressure which then decreases cerebral perfusion pressure
- Chronicity matters. In chronic respiratory acidosis, the kidneys help balance the blood acidity, which reduces the impact of high CO₂ levels in the CSF. This makes chronic respiratory acidosis less symptomatic than acute respiratory acidosis.
- Patients will also present with symptoms of the underlying driver of low respiratory ventilation itself (dyspnea from COPD, muscle weakness from myasthenia gravis, pinpoint pupils from opioid overdose, etc.)

Next, we mentally subdivide patients into **acute and chronic presentations**. First, patients may have an **acute** presentation due to a sudden event like a cardiac arrest or a drug overdose. These patients are more likely to have altered mental status due to a low pH level and are more likely to be seen in the midst of a COPD exacerbation with significant dyspnea. In contrast, other patients with chronic muscular, CNS, or lung disease will present with **chronic** respiratory acidosis, which is more or less incidentally discovered on laboratory testing in a mostly asymptomatic patient.

 CLINICAL CORRELATION

A common presentation for acute respiratory acidosis in the hospital is for patients to be found with altered mental status in the middle of the night. An arterial blood gas will often show acidosis and hypercapnia (a high PCO_2).

What Is the Pathophysiology of Respiratory Acidosis? (LO2)

The body's pH is tightly regulated between **7.35 and 7.45**, a narrow range essential for normal cellular function. From a chemical perspective, pH reflects the concentration of hydrogen ions ($[\text{H}^+]$) in body fluids. The relationship between pH and $[\text{H}^+]$ is logarithmic, so each 0.3 change in pH corresponds to a doubling or halving of $[\text{H}^+]$. To manage keep the pH within a narrow range, the body uses two main systems:

Volatile acids (mainly carbon dioxide, CO_2)

- produced continuously by the Krebs cycle in aerobic metabolism.
- CO_2 combines with water to form carbonic acid (H_2CO_3), which dissociates into hydrogen ions (H^+) and bicarbonate (HCO_3^-).
- CO_2 is eliminated by the lungs through ventilation.

Non-volatile acids (e.g., sulfuric, phosphoric, lactic acids)

- Generated from protein metabolism and other metabolic processes.
- Cannot be excreted via the lungs, so they are removed by the kidneys, which secrete H^+ and reabsorb/regenerate HCO_3^- .

During rapid fluctuations in acid levels throughout the day, the body needs a system to quickly neutralize acids to keep the pH stable. The **bicarbonate**

buffer system is the body's primary extracellular buffer and is represented by:

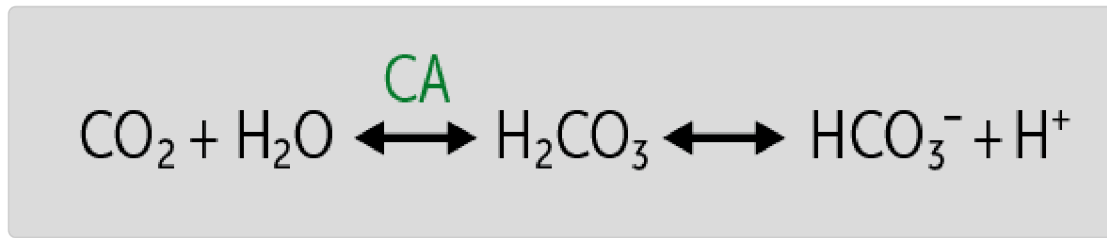


Figure 1

Changes in ventilation alter CO_2 levels, which shift this equation and the body's pH.

- **Hypoventilation** $\rightarrow \uparrow \text{CO}_2 \rightarrow \uparrow \text{H}^+ \rightarrow \downarrow \text{pH}$ (respiratory acidosis)
- **Hyperventilation** $\rightarrow \downarrow \text{CO}_2 \rightarrow \downarrow \text{H}^+ \rightarrow \uparrow \text{pH}$ (respiratory alkalosis)

In normal physiology, ventilation matches metabolic CO_2 production to keep PaCO_2 near 40 mmHg. Respiratory acidosis occurs when alveolar falls below metabolic demand, causing **excessive CO_2 retention** (hypercapnia) and a **fall in pH**. The disturbance originates in the respiratory system, and compensation occurs through the kidneys.

Etiology

PaCO_2 will be high if there are problems with CO_2 excretion by the lungs. During normal physiology, CO_2 excretion depends on adequate ventilation (a function of respiratory rate and tidal volume) as well as adequate gas exchange across the alveolar membrane (CO_2 must be able to move from blood into the alveoli for expiration). Therefore, the two major mechanisms

of respiratory acidosis are **impaired ventilation** (rate and/or tidal volume) and **impaired gas exchange**. Some examples are below.

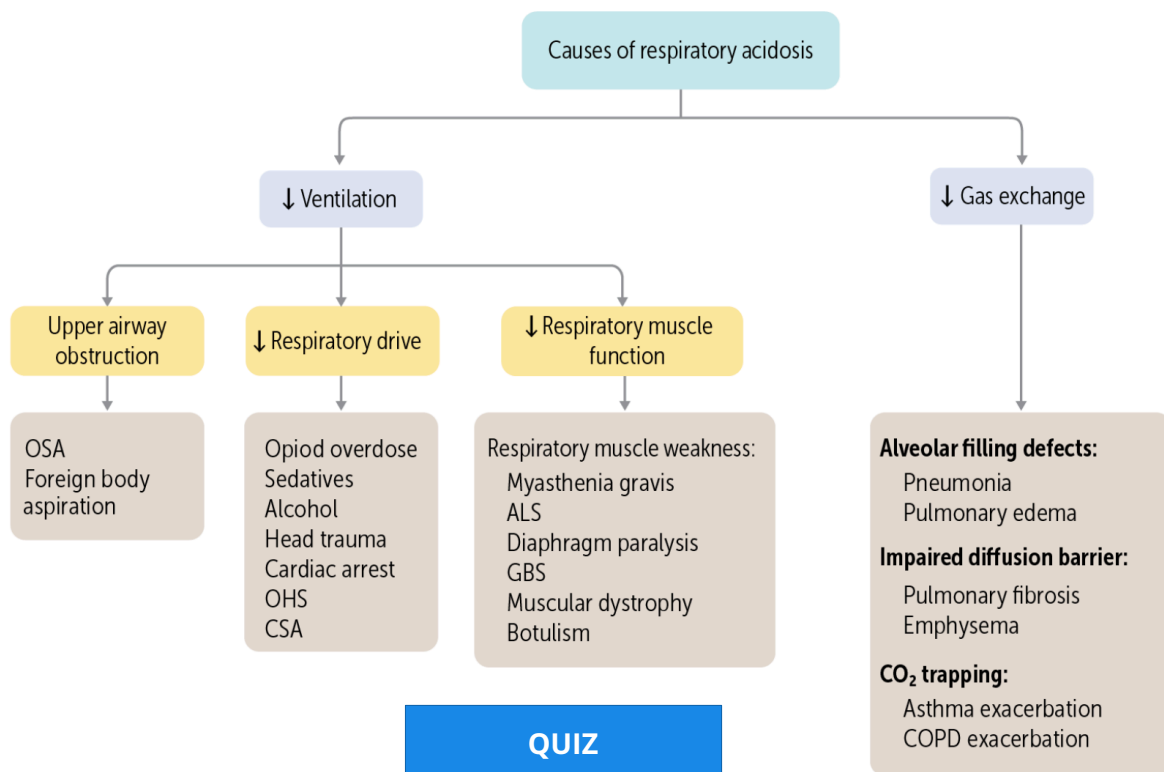
Impaired Ventilation. Here, breathing is slow and/or weak. Some main causes are:

- Decreased respiratory drive: Opioids, sedatives, trauma, alcohol, cardiac arrest, obesity hypoventilation syndrome (OHS), and central sleep apnea (CSA) can all cause decreased respiratory drive, with decreased respiratory rate and/or tidal volume.
- Decreased respiratory muscle function: This is impaired in neuromuscular diseases such as myasthenia gravis, amyotrophic lateral sclerosis (ALS), diaphragm paralysis, Guillain-Barré syndrome (GBS), muscular dystrophy, and botulism, leading to reduced tidal volume.
- Decreased ventilation due to obstruction: Tidal volume can also be decreased by upper airway obstruction in obstructive sleep apnea (OSA) and foreign body aspiration.

Impaired Gas Exchange. The lungs are diseased, preventing CO₂ from moving across the alveolar membrane. Some causes are:

- Pneumonia and pulmonary edema, which impair gas exchange by filling alveolar spaces with exudates.
- Emphysema, which decreases the alveolar surface area available for gas exchange.
- COPD and asthma exacerbations, which cause airway obstruction that traps CO₂ within the lungs, which diffuses back into the blood. These disorders will also decrease ventilation.

A common reason for hospital admissions are COPD exacerbations. These patients present with cough, shortness of breath, and an acute respiratory acidosis overlying a chronic respiratory acidosis. They are initially treated with bronchodilators, but if they have ongoing respiratory acidosis that does not improve with medications, bilevel positive airway pressure (BPAP) is often initiated.



Compensation for Respiratory Acidosis

In respiratory acidosis, the blood gas findings follow a characteristic pattern:



In **respiratory acidosis**, the initial event is a high CO_2 , and the equilibrium shifts towards the right — so the body is in a **hydrogen-rich state**.

The **kidneys respond** by:

- **Excreting H^+** into the urine, buffered by phosphate and ammonium.
- For each H^+ excreted, the kidneys are able to ***regenerate a new bicarbonate*** and add it back into the blood.

That extra bicarbonate is now additional buffering capacity and essentially acts like a **magnet for free protons**. So, even though CO_2 is still high, the Henderson–Hasselbalch equation balance (**Figure 1**) improves as the bicarbonate increases enough to rebalance the pH.

In other words, the kidneys **remove H^+ from the system to help maintain a stable pH**. From the equation's perspective, this pushes the balance back toward the bicarbonate side and prevents runaway acidification. The longer the kidneys have to accomplish this, the higher they increase the bicarbonate level. As such, higher bicarbonate levels are seen in **chronic** respiratory acidosis.

In contrast, in **acute** respiratory acidosis (eg, due to opioid overdose or cardiac arrest) there is no time for this compensation, so the serum bicarbonate concentration may be normal or just minimally increased by 1 to 2 mEq/L. The quantitative degree of expected compensation will be discussed more in the Acid-Base Disorders: Putting It All Together brick.

respectively.

bicarbonate (HCO_3^-) as the weak acid and conjugate base,

The bicarbonate buffer system uses carbonic acid (H_2CO_3) and

How Do We Diagnose Respiratory Acidosis? (LO3)

Respiratory acidosis should be considered in any patient with altered mental state, or with pulmonary symptoms, especially if the respiratory rate is reduced.

Physical Examination

Altered mental status, abnormally slow or shallow breathing, pursed lip breathing, or wheezing may be seen on physical exam. If any of these are present, consider the possibility of respiratory acidosis.

Laboratory Tests

Use an arterial blood gas analysis and the basic metabolic panel to identify the expected pattern of decreased pH, increased PaCO_2 , and increased bicarbonate (the latter only if the disorder is chronic).

Other Tests

Use other diagnostic testing specific to the suspected underlying cause. For example, a chest x-ray or CT scan is needed in suspected pulmonary disease. Sleep studies for sleep apnea or pulmonary function testing to confirm COPD may be necessary, depending on the patient's presentation.

diagnose the cause of respiratory acidosis.

Chest x-ray or CT may be helpful, depending on the patient history, to

How Do We Manage Respiratory Acidosis? (LO4)

An important cause of respiratory acidosis to recognize is **opioid overdose**. Potential cases of this (evidenced by slow, shallow respirations, mental state change, pinpoint pupils) should receive the opioid antagonist **naloxone**, even before the complete workup is done.

CLINICAL CORRELATION

Since patients with an acute respiratory acidosis are often confused, it is difficult to get an accurate history of opioid use from them. Naloxone is a very safe medication that is often given if there is suspected opioid use. Patients will begin to wake up within minutes of giving naloxone if they have taken opioids.

Otherwise, management of respiratory acidosis depends on the cause, but the overall goal is to improve ventilation. Bilevel positive airway pressure (BiPAP) is often used to increase the depth of ventilation (tidal volume) and help “blow off” CO_2 , depending on the cause of respiratory acidosis.

Patients with markedly decreased partial pressure of oxygen, rising PaCO_2 , or who continue to become fatigued despite continuous positive airway pressure (CPAP)/BiPAP require endotracheal intubation, as these findings are preludes to respiratory arrest. These maneuvers are often done even prior to a final diagnosis.

suspected.

before a complete workup is done when an opioid overdose is suspected, an opioid antagonist, should always be administered even

CASE CONNECTION

[BACK TO INTRODUCTION](#) 

Thinking back to WN, what is your diagnosis? What are your immediate first steps?

You recognize opioid overdose. You obtain an arterial blood gas while the nurse immediately administers naloxone, an opioid antagonist. WN's arterial blood gas shows elevated PaCO_2 due to hypoventilation and a low pH: acute respiratory acidosis. After receiving naloxone, WN's respiratory rate increases, and he slowly becomes alert. You sit with him and discuss his close brush with death. You explain that when he was brought in, his carbon dioxide levels were high because of his low respiratory rate. When the lungs are unable to remove carbon dioxide, the blood becomes too acidic, and that can lead to death. You tell him that because of his friend's quick actions, he saved his life.

Summary

- Respiratory acidosis is a primary acid-base disorder with elevated partial pressure of arterial carbon dioxide (PaCO_2) and reduced pH.
- Initial presentation may be with altered mental state or decreased respiratory rate.
- Other findings are specific to the underlying cause.
- Respiratory acidosis is caused by either decreased pulmonary gas exchange or decreased ventilation (eg, low respiratory rate or tidal volume).
- Causes include central (CNS) depression, weak respiratory muscles, upper airway obstruction, and lung diseases that impair alveolar gas exchange.
- Kidneys compensate for respiratory acidosis by retaining bicarbonate, raising the pH toward normal in chronic respiratory acidosis.
- The diagnosis of respiratory acidosis is confirmed by a venous or arterial blood gas.

- Chest radiography, or more specialized tests like CT and pulmonary function testing, may be helpful for discovering the underlying cause.
- Naloxone should be given in suspected opiate overdose without delay.
- Bilevel positive airway pressure (BiPAP) can increase tidal volume in many disorders with low tidal volume.

Review Questions

1. A 23-year-old female is brought to the emergency department somnolent and minimally responsive to painful stimulation. Her exam is notable for a respiratory rate of 8 breaths/min. Her heart and lung exams are normal. Her arterial blood gas shows the following: pH 7.30; partial pressure of arterial carbon dioxide, 55 mm Hg; bicarbonate, 26 mEq/L; and partial pressure of oxygen, 85 mm Hg. Which of the following diagnoses is most likely?

- A. Asthma
- B. Myasthenia gravis
- C. Obstructive sleep apnea
- D. Pneumococcal pneumonia
- E. Sedative overdose

Explanation (requires correct answer)

2. Which of the following is most helpful in initially diagnosing respiratory acidosis?

- A. Chest x-ray
- B. Pulmonary function testing
- C. Pulse oximetry
- D. Serum electrolytes
- E. Venous blood gas

Explanation (requires correct answer)

3. Which of the following best describes the renal compensation for respiratory acidosis?

- A. The blood pH becomes alkalemic.
- B. The blood pH corrects to 7.40.
- C. The serum bicarbonate falls, more in acute disease.
- D. The serum bicarbonate falls, more in chronic disease.
- E. The serum bicarbonate rises, more in acute disease.
- F. The serum bicarbonate rises, more in chronic disease.

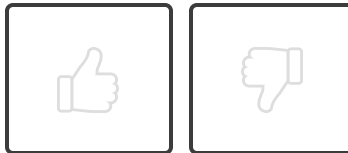
Explanation (requires correct answer)

4. You are on call in the hospital and are called to a room at midnight. You find a somnolent patient with pinpoint pupils and a respiratory rate of 9 breaths/min. They are a chronic kidney disease patient who recently had abdominal surgery. They have been receiving IV morphine for pain control. You suspect that the renally-cleared morphine has risen in concentration in the blood and that they may have respiratory acidosis. An arterial blood gas and accompanying basic metabolic panel confirm this diagnosis. What will provide definitive management of the cause of their respiratory acidosis?

- A. Intravenous sodium bicarbonate
- B. Emergency dialysis
- C. Naloxone
- D. Bilevel positive airway pressure (BiPAP)

Explanation (requires correct answer)

How would you rate this Brick?



Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01 Differentiate the clinical presentation of respiratory acidosis.
- 02 Explain the pathophysiology of respiratory acidosis.
- 03 Interpret clinical and lab findings to properly diagnose respiratory acidosis.
- 04 Outline the management plan for respiratory acidosis.

Objective Confidence: 0/4